Qingyang Zhou

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https://esperarr.github.io/CV/ ⋄ https://github.com/Esperarr

EDUCATION

Tsinghua University September 2022 – Expected 2027
Xinya College & Undergraduate Program in Computer Science and Technology
GPA: 3.6/4.0
Selected coursework: Complex Analysis (A+), Media Computing (A+), Database Special Topic Training (A+), Introduction to Artificial Intelligence, Introduction to Machine Learning, Fundamentals of Computer Graphics.

MANUSCRIPTS

"Curriculum Multi-Reward Reinforcement Learning for Customized Text-to-Image Generation"
Yuwei Zhou, Xin Wang, Hong Chen, Yipeng Zhang, ***Qingyang Zhou****, Wenwu Zhu*
Manuscript OpenReview Paper PDF

"Towards Mutually Illuminative Collaborative Human–AI Deliberate Discussion"
Zipeng Zhang, Shiwei Wu, Ran Ran, ***Qingyang Zhou****, Ranrui Ma, Jiaye Leng, Jian Zeng, Zhenhui, Chun Yu*
Manuscript Paper PDF

RESEARCH EXPERIENCE

Research Intern, Information Retrieval Lab October 2024 – July 2025
Tsinghua University Information Retrieval Lab

- Surveyed recommendation-model literature and reproduced open-source models (primarily FM and DeepFM), studying their implementations.
- Integrated an existing sparse autoencoder (SAE) plug-in into the codebase and ran preliminary experiments under a senior student's guidance.

Research Intern, Pervasive Human-Computer Interaction Lab January 2025 – May 2025
Tsinghua University Pervasive Human-Computer Interaction Laboratory

- Surveyed and presented literature on aligning human and AI thought processes; contributed to group discussions on the research direction.
- Implemented part of the core logic for a human–AI interactive writing assistant that supports collaborative idea generation.
- Co-authored "Towards Mutually Illuminative Collaborative Human–AI Deliberate Discussion" (see Manuscripts).

Research Intern, Multimedia and Network Big Data Lab July 2023 – July 2024
Tsinghua University Multimedia and Network Big Data Lab

- For the project "Video Generation System Combining LLMs with Diffusion Models," reproduced experiments from multimodal LLM papers and presented results at group meetings.
- Assisted senior students in debugging experimental code; studied diffusion models and personalized image generation.
- Co-authored "Curriculum Multi-Reward Reinforcement Learning for Customized Text-to-Image Generation" (see Manuscripts).

INDUSTRY EXPERIENCE

Client Development Engineer Intern, Tencent (Project MAX) May 2026 – August 2026
Tencent, Project MAX Team

- Improved NavMesh-based character movement in battle, resolving stuttering on terrain bumps and sticking at convex corners; fixed characters leaving the map during dash displacement by constraining destinations to the navigable region (boundary containment test with nearest-walkable-point sampling).
- Built a Scene-view tool for diagnosing movement anomalies in the fixed-point logic layer: it reads runtime state via reflection and renders target points, input direction, and trajectories, with automatic stuck detection.
- Developed in-battle GM tools (performance overlays for 2V2 mode; a buff inspector for the test scene), the captain-

selection hero showcase (model fallback, entrance-to-idle blending, camera zoom-and-lock), and a priority-based main-city interaction-prompt system; documented the NavMesh baking workflow.

SELECTED PROJECTS

Path Tracing Renderer [Source code] June 2026

Computer Graphics Course Project

- Developed a physically based renderer supporting Whitted-style ray tracing and Monte Carlo path tracing with Russian roulette termination.
- Implemented physically based materials with GGX microfacet importance sampling (isotropic and anisotropic), the Cook–Torrance BRDF, and Schlick's Fresnel approximation, covering diffuse, mirror, glass, and glossy surfaces.
- Added BVH acceleration for meshes with over 100k faces, next-event estimation (NEE) with multiple importance sampling (MIS, power heuristic), depth of field, MSAA, gamma correction, and OpenMP parallelization.

Five-Stage Pipeline CPU [Source code] November 2025 – December 2025

Computer Organization Course Project

- Implemented a five-stage pipelined RISC-V CPU in SystemVerilog covering 19 basic instructions plus extensions, with data forwarding, control-signal generation, and hazard handling; validated through Vivado simulation.

Connect Four AI [Source code] June 2024

Course Project

- Developed a Connect Four AI in Python using Monte Carlo Tree Search (MCTS) with Upper Confidence bounds applied to Trees (UCT), achieving a win rate above 90% against baseline AI opponents.

Text Sentiment Analysis Model [Source code] May 2024

Course Project

- Implemented and compared CNN, RNN, and MLP text sentiment classifiers in PyTorch on a course-provided dataset, achieving over 90% accuracy.

Intelligent Agent-Based Game "Generals" September 2023 – May 2024

Major Project in the 28th Tsinghua University Intelligent Agent Competition

- Lead developer of a turn-based strategy game for the competition, available at saiblo.net/game/35.
- Responsible for backend logic, unit testing, debugging, SDK development, and documentation; supported participants throughout the competition.

TEACHING AND SERVICE

Volunteer Teaching Assistant (Programming and Algorithms) September 2025 – Present

Tsinghua University, Center for Student Learning and Development

- Provide drop-in tutoring in C/C++, data structures, and algorithms, covering debugging, algorithm design, and core concepts.

Secretary, Intelligent Agent Department February 2023 – Present

Tsinghua CS Student Science and Technology Association

- Participate in game development and event organization for the Tsinghua Intelligent Agent Competition; led development of "Generals" for the 28th edition (see Projects).

Tutor, Tsinghua University Science and Engineering Summer Camp July 2024 – August 2024

2024 Science and Engineering Summer Camp, I6 Class Tutor

- Provided technical guidance and mentored teams in intelligent robot design and cardboard-bridge construction.

SKILLS

Languages	Chinese (native), English (TOEFL 108), Spanish (basic)
Programming	Python, C/C++, C#, SystemVerilog, Java, JavaScript/TypeScript, Rust
AI/Machine Learning	PyTorch, NumPy, diffusion models, recommendation models, deep learning
Graphics/Game Development	OpenGL, Unity, path tracing, rendering pipelines, NavMesh pathfinding
Tools	Git, Linux, Docker, LaTeX, Markdown